

Bonding triangle

Learning outcomes

By the end of this section you should be able to:

- Determine the position of a compound based on its position in the bonding triangle from electronegativity data.
- Predict properties of a compound based on its position in the bonding triangle.

Chemists have long desired the ability to classify substances based on their bond types to explain physical properties. As we saw in [section S2.4.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/>) chemical bonding does not exist in discrete categories but rather as a continuum between ionic, covalent and metallic models. Anton van Arkel and Jan Ketelaar were two chemists that attempted to classify bonds along a bonding continuum by adding more detail to the simple bonding triangle that we looked at earlier. **Interactive 1** below shows two examples of these bonding triangles. Scroll through the photos to view them.

Interactive 1. Van Arkel's 1941 Bonding Triangle

 More information for interactive 1

The slide 1 in the interactive is Van Arkel's bonding triangle, a model developed to classify different types of chemical bonds based on the properties of elements and compounds. This triangle is divided into three main bonding types: ionic bond (top of the triangle), covalent bond (bottom left corner) and metallic bond (bottom right corner).

Ionic Bonds — Formed when electrons are completely transferred from one atom to another, creating charged ions. Example mentioned in the triangle: CsF (Caesium Fluoride).

Covalent Bond — Formed when atoms share electrons to achieve stability. Examples mentioned in the triangle: F_2 (fluorine gas) and I_2 (iodine gas).

Metallic Bonds — Found in metals, where electrons move freely among positively charged ions. Example mentioned in the triangle: Sn (Tin).

Compounds along the edges show a mix of bonding characteristics. BF_3 , CF_4 , and NF_3 lie between ionic and covalent bonds. AlP , $ZnTe$, and $SnPb$ are mentioned between metallic and covalent bonds. CoS is between ionic and metallic bonds, showing both properties.

This diagram helps students understand bonding variations and bond types, rather than treating them as separate categories.

Slide 2 interactive displays Ketelaar's bonding triangle, a model that classifies chemical bonds based on the properties of different compounds. The triangle is divided into three main bonding categories:

Metallic Bonds (Top Corner) — Found in metals, where electrons move freely. Example mentioned in the triangle: Li (Lithium), Ag (Silver), Sn (Tin).

Covalent Bonds (Bottom Left Corner) — Occur when atoms share electrons to form stable molecules. Examples mentioned in the triangle: F_2 (Fluorine), I_2 (Iodine), ClF (Chlorine monofluoride).

Ionic Bonds (Bottom Right Corner) — Formed when electrons are transferred between atoms, creating charged ions. Example: CsF (Caesium).

This diagram allows users to see that bonding falls on a spectrum that includes covalent, ionic, and metallic bonds.

The examples of bonding triangles shown in **Interactive 1** is a way to classify specific bonds. While such diagrams are useful, they are an estimate of the relative location of a substance on the bonding continuum, based on observed properties. Can you think of any ways to quantify a diagram of this type? Do you know of any numerical parameters that describe bond characteristics?

When we considered the bonding continuum earlier in [section S2.4.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/>) we looked at how polar covalent bonds are a bond type between ionic and covalent. What numerical parameter can you use to deduce the polar nature of a covalent bond? Recall from [section S2.2.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>) that electronegativity is the tendency of an atom to attract a shared pair of electrons. Each atom has an assigned electronegativity value, relative to fluorine, that can be found in [section 9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet. These values provide an indicator about the degree to which an individual atom will attract an additional electron from another atom.

- Metals have a high tendency to lose electrons, low tendency to attract additional electrons, therefore low electronegativity values.
- Non-metals have a high tendency to gain electrons, therefore high electronegativity values.

How might we use electronegativity values to quantify bond types between two elements? Consider the types of elements and the electronegativity values from each element for each bond type. **Table 1** shows a summary of the types of elements and electronegativities for each bond type.

Table 1. Types of elements and electronegativities for the bond type between two elements

Bond type	Types of elements	Electronegativity	
Ionic	Metal	Low	Large difference in electronegativity
	Non-metal	High	
Covalent	Non-metal	High	High average electronegativity
	Non-metal	High	
Metallic	Metal	Low	Low average electronegativity
	Metal	Low	

Nature of Science

Aspects:

- Models
- Science as a shared endeavour

The original bonding triangles were qualitative until University of Cincinnati professor William B. Jensen began quantifying van Arkel's and Ketelaar's triangles of bonding using electronegativity values for the purposes of teaching his own students. He published a quantified triangle in the 1980s and again in the 1990s, but his name is frequently omitted from the bonding triangle. Different techniques and mathematical analyses are currently being used to further develop bonding triangles by universities around the world. The most common one currently used is the Van Arkel-Ketelaar triangle. You can find a version of this triangle in [section 17 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/) of the DP chemistry data booklet and shown below in **Figure 1**.

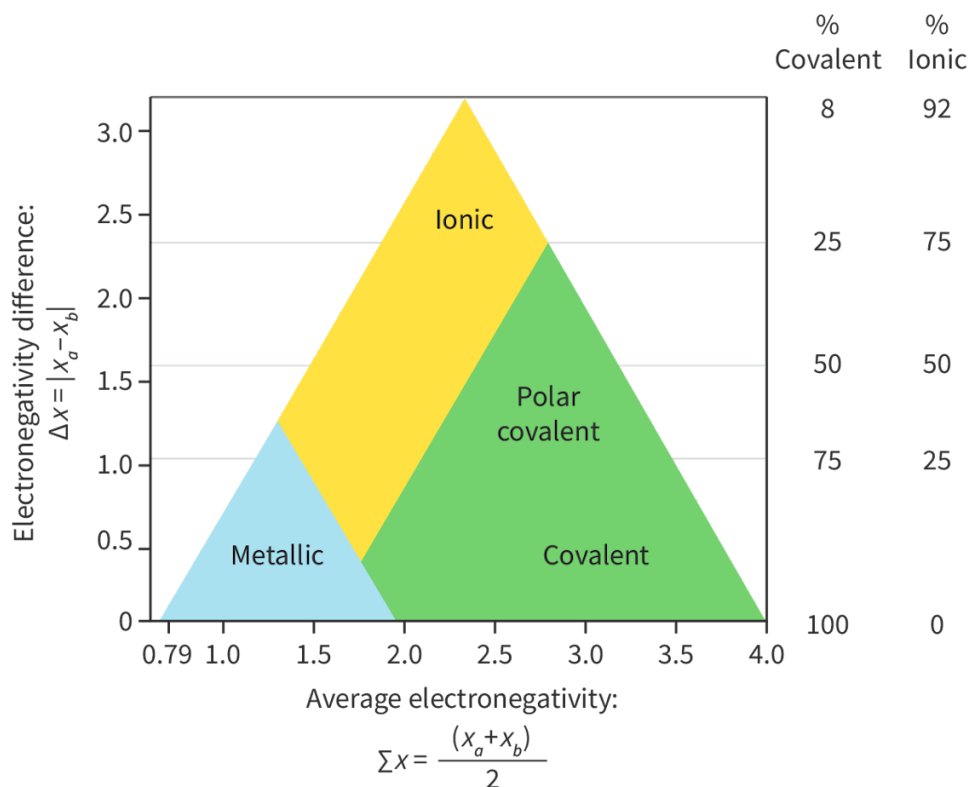


Figure 1. van Arkel-Ketelaar triangle of bonding diagram.

More information for figure 1

The image is a diagram known as the van Arkel-Ketelaar triangle, used to illustrate the types of chemical bonding between two elements based on their electronegativity. The x-axis represents the average electronegativity $((x_a + x_b) / 2)$ ranging from 0.79 to 4.0, while the y-axis represents the difference in electronegativity $(\Delta x = |x_a - x_b|)$ ranging from 0 to 3.0.

The diagram is a triangle divided into three regions: "Metallic" at the bottom-left with lower values of both average and difference in electronegativity, "Polar covalent" in the middle, and "Ionic" at the top with higher values. To the right of the triangle, there are percentages indicating the covalent and ionic character ranging from 8% covalent and 92% ionic to 100% covalent and 0% ionic.

[Generated by AI]

This version quantifies the bonding triangle by plotting the average electronegativity between two elements on the x-axis and the difference of electronegativity between the two elements on the y-axis. Where the point falls within the triangle would be its bond type. While the distinctions between the three bond types are fairly clear on the diagram, the distinction between pure covalent and polar covalent is not. This is where deducing the percentage covalent and ionic character can be useful. You will notice these percentages provided on the right-hand side of the bonding triangle in **Figure 1**.

Let's consider where sodium, hydrogen and sodium chloride would fall on the bonding triangle. We would first need to start by identifying the electronegativities for each element followed by calculating the average electronegativity and the difference in electronegativity. This is shown below in **Table 2**.

Table 2. Electronegativities, difference in electronegativities and average sodium, hydrogen and sodium chloride.

	Na	H ₂
Electronegativity χ_a and χ_b	Sodium 0.9	Hydrogen 2.2
	Bonding occurs between atoms of Na	Bonding occurs between atoms of H
Difference in electronegativity	$\Delta\chi = \chi_a - \chi_b $ $\Delta\chi = 0.9 - 0.9$ $= 0$	$\Delta\chi = \chi_a - \chi_b $ $\Delta\chi = 2.2 - 2.2$ $= 0$
Average electronegativity	$\sum \chi = \frac{(\chi_a + \chi_b)}{2}$ $\sum \chi = \frac{(0.9 + 0.9)}{2}$ $= 0.9$	$\sum \chi = \frac{(\chi_a + \chi_b)}{2}$ $\sum \chi = \frac{(2.2 + 2.2)}{2}$ $= 2.2$
Bond type see Figure 2	Metallic	Covalent

You will notice that for sodium and hydrogen, the bonding occurs between atoms of the same element, resulting in a difference in electronegativity of zero. Both sodium and hydrogen would be plotted on the x-axis of the bonding triangle and have a 0% ionic character as shown in **Figure 2**.

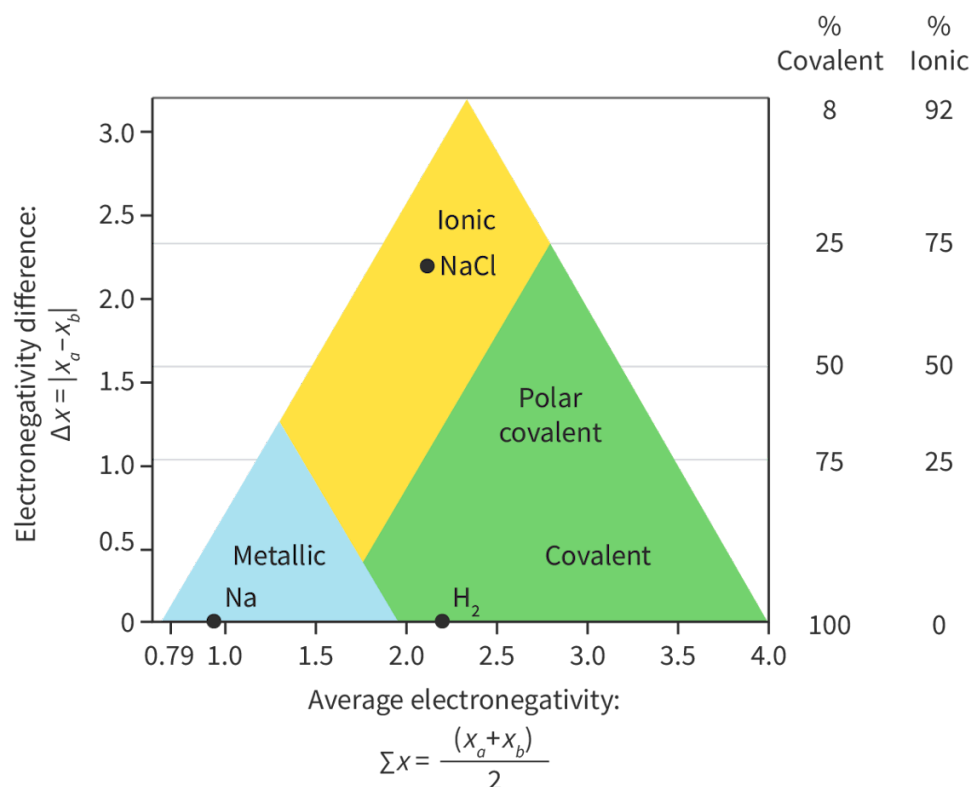


Figure 2. Bonding triangle with Na, H₂ and NaCl plotted onto it.

[More information for figure 2](#)

The image is a bonding triangle graph plotting different compounds based on their electronegativity properties. The X-axis represents the average electronegativity labeled as " $(x_a + x_b)/2$," ranging from 0.79 to 4.0. The Y-axis represents the electronegativity difference, labeled as " $|x_a - x_b|$," ranging from 0 to 3.0. The triangle is divided into three distinct regions.

1. The bottom left blue area is labeled "Metallic" with "Na" marked there.
2. The top yellow section is labeled "Ionic" with "NaCl" marked there.
3. The right green section is divided into "Polar covalent" and "Covalent" subregions, with "H₂" plotted in the "Covalent" area.

Percentages at the right show covalent and ionic character, with 100% covalent at the bottom and 100% ionic at the top.

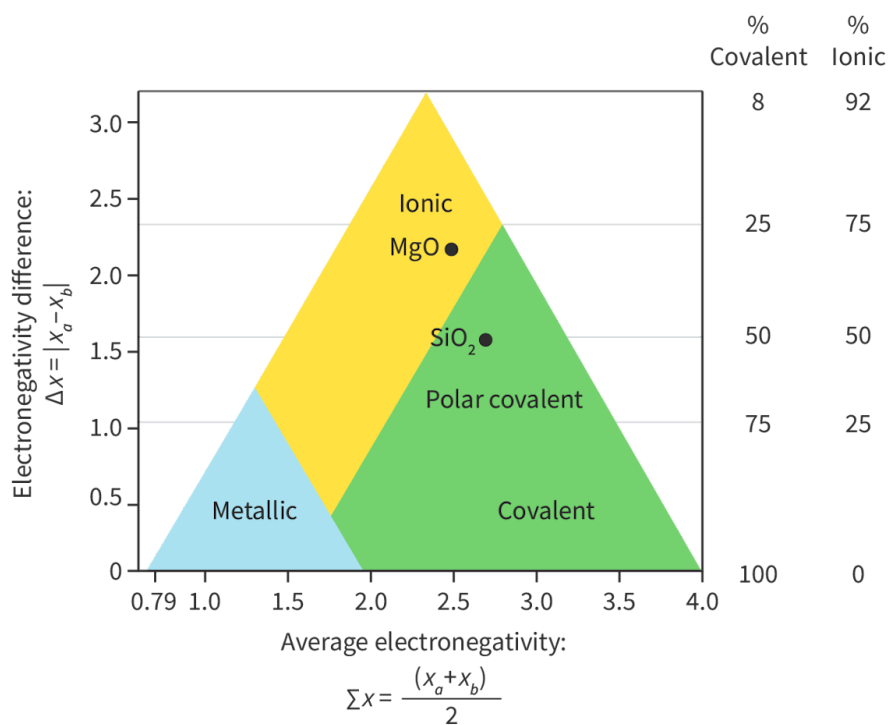
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Let's try another example.

Worked example 1

Silicon dioxide and magnesium oxide are both found within the Earth's crust. Classify the bond type and estimate the percent covalent character for each.

	SiO₂	
Electronegativity χ_a and χ_b	Oxygen 3.4	
	Silicon 1.9	
Difference in electronegativity	$\Delta\chi = 3.4 - 1.9$ $= 1.5$	
Average electronegativity	$\sum\chi = \frac{(3.4 + 1.9)}{2}$ $= 2.65$	
Bond type see diagram below	Covalent	
Percent covalent character	Approximately 50%	



The physical properties of elemental substances, metals and various binary compounds can be explained by considering the structure, bonding and the position in the bonding triangle.

Study skills

It is important to note that we cannot use the bonding triangle for substances made up of more than two elements (e.g. ionic compounds with polyatomic ions and molecules with more than two different atoms). It can only be used for elemental substances and binary compounds.

In the drag and drop interactive, drag the appropriate substances (A–H) into their respective locations in the bonding triangle.

Interactive 2. Where do the Seven Substances go in the Bonding Triangle?

 More information for interactive 2

This interactive image displays the **van Arkel—Ketelaar triangle**, a tool used to classify the type of bonding between elements based on electronegativity values.

The diagram plots the electronegativity difference: $\Delta\chi = |\chi_a - \chi_b|$ on the y-axis and average electronegativity: $\Sigma\chi = \frac{(\chi_a + \chi_b)}{2}$ on the x-axis. The x-axis ranges from 0.80 to 4.0, and the y-axis ranges from 0 to 3.2.

The triangle is divided into **three colored regions** representing different bonding types:

- **Yellow (top left): Ionic bonding**
- **Green (lower right): Polar covalent bonding**

- **Blue (lower left): Metallic bonding**

Users must drag and drop the correct molecules and elements into the corresponding positions in the diagram. The drop zones are provided at the following locations:

Top right of the yellow section

The bottom right of the yellow section

The top center of the green section

The bottom right of the green section

The bottom left corner of the green section

The bottom right corner of the blue section

The bottom left of the blue section

The drag-and-drop options provided are:

A- Magnesium hydride

B - Potassium

C - Water

D - Sodium chloride

E - Oxygen

F - Aluminium

G - Silicon

This interactivity helps users understand how electronegativity differences and averages determine the type of chemical bonding.

Read below for the solution:

Top right of the yellow section - D - Sodium chloride

The bottom right of the yellow section - A- Magnesium hydride

The top center of the green section - C - Water

The bottom right of the green section - E - Oxygen

The bottom left corner of the green section - G - Silicon

The bottom right corner of the blue section - F - Aluminium

The bottom left of the blue section - B - Potassium

This activity helps learners understand how **bond type correlates with electronegativity difference and average electronegativity:**

Theory of Knowledge

We use scientific models to simplify theories and to make predictions. You have just learned how to use the van Arkel-Ketelaar bonding triangle to predict the bonding in binary compounds as metallic, ionic or covalent. The concept of a bonding triangle has persisted in chemistry and many variations have been developed and modified over time. It is possible that in time, the van Arkel-Ketelaar triangle will undergo another change with advancements in knowledge and experimental evidence. Can the use of a changing scientific model reliably predict the characteristics of all compounds?

Now that you have identified the bond types of each of the substances (A–G) drag and drop the associated structure and notable characteristic that corresponds to each substance in this interactive task.

Interactive 3. Identify the Bond Type, Structure and Notable Properties of the Seven Substances.

 More information for interactive 3

What might you expect about the bonding and properties of substances that fall along a boundary line? Since bonding is best described as a continuum, substances that fall along boundary lines show intermediate bonding and properties. For example, covalent network solids tend to fall near the metallic and

ionic boundaries but still within the covalent region of the triangle. These substances have lattice structures and high melting points like metals and ionic compounds, while still forming covalent bonds between atoms.

SiO₂ is an example of a substance we looked at in the worked example that is in the covalent region but falls close to the ionic bonding boundary line. SiO₂ is an example of a network covalent structure we explored in [section S2.2.7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/>) that is brittle, has a high melting point (1700 °C), but still has covalent bonds between atoms of silicon and oxygen.

In this activity, you will have the opportunity to think about how all three types of bonding are important for building structures out of reinforced concrete.

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication skills — Using digital media for communicating information
- **Time required to complete activity:** 45 minutes
- **Activity type:** Pair activity

Reinforced concrete is a composite material that is an example of a heterogeneous mixture. The different components of a composite material each have different properties that complement one another when combined. It is one of the most widely used building materials globally. The most common substances found in reinforced concrete are:

Component Type	Composition
Concrete	Cement (primary made up of C
	Rocks and aggregates to make concrete (SiO ₂ , Fe ₂ O ₃)
Reinforcement	Rebar (primarily made

Complete the following with a partner:

- Assign where each of the substances would fall in the triangle of bonding.
- Estimate the percent ionic character for each substance.
- Discuss with your partner what properties you would expect each substance to have based on the position within the bonding triangle.
- Select a building structure of your choosing from your local community built from reinforced concrete. This could be a bridge, building, dam, pier, stadium, etc.

- Create a piece of media (video commercial, magazine advertisement, podcast, poster) explaining how the atomic structure and bonding of each substance gives rise to the desired properties of your building structure.